

Evaluating the Impact of Sonification in an Immersive Analytics Environment Using Real-World Geophysical Datasets

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Figure 1: User study Task 1 visualization (left), and Task 2 visualization (right), both used to analyze different real-world space weather datasets in a mixed reality environment.

ABSTRACT

In this paper, we evaluated the impact of audio in an immersive analytics environment using real-world geophysical datasets. We specifically used sonification, i.e., the use of non-speech audio to convey information. To evaluate the impact of sonification, we designed a between-subject experiment in a mixed-reality environment and conducted a user study with 50 participants in two scenarios: Audio-visual and Visual-only. In the study, we compared task metrics such as the number of patterns identified by the participants, the level of confidence, participants' task responses, the NASA task load index, and the SUS questionnaire between each scenario to study the role of sonification in augmenting the analysis process in an immersive environment. We found that event-based sonification, used in addition to the visual channel, is helpful in finding patterns and relations in geophysical datasets. Our results also suggest that using audio in immersive analytics might increase users' confidence in performing analytics tasks such as pattern finding. Using real-world datasets, we identified the advantages and limitations of using sonification in an immersive analytics context.

Index Terms: Immersive analytics, sonification, human-centered design, mixed reality environment, spatiotemporal data, frame of reference.

1 INTRODUCTION

Immersive analytics is an emerging form of data analysis and decision-making that leverages immersive environments and human embodiment to perform data analytics. Despite advances in visualization, the integration of sonification, i.e., the use of non-speech audio to convey information [1], remains underexplored. This study evaluates the impact of sonification on user understanding of geophysical datasets in immersive environments. The study uses real-world space weather data to investigate whether com-

binning sonification with visualization improves pattern recognition, confidence, and task performance.

2 METHODOLOGY

Using Microsoft HoloLens 2, participants explored geophysical space weather datasets in two scenarios, audio-visual and visual-only. The study utilized three real-world datasets:

SYM-H Index: The SYM-H index measures the intensity of a solar storm. More negative SYM-H indices correspond to more intense solar storms. Figure 1 (left) depicts the variation in the SYM-H index during a solar storm, plotted over 48 hours duration.

Sunspot Number/ Solar Activity Level: The sunspot number is an empirically measured positive integer that has long been used as a proxy for the overall solar activity level.

Total Electron Content (TEC): TEC is the total number of electrons in the upper atmosphere present along a path between a radio transmitter and receiver. The TEC is modified by changing solar Extreme UV radiation, solar storms, and atmospheric waves. For ground-to-satellite communication and satellite navigation, TEC is a good parameter to monitor for possible space weather impacts.

2.1 Sound Design

The immersive system incorporated three types of sound:

1. **Event-based audio:** We used natural sounds to indicate sunrise and sunset events. As participants dragged the slicing plane across the time series line plot and crossed the sunrise time, the rooster audio played, marking that the sunrise had happened. Similarly, to indicate sunset, we used the sound of cicadas.
2. **Action-based audio:** The sound associated with the interactive slicing plane was designed to respond dynamically to user interaction. When a user interacts with the slicing plane, the sound mapped to the data emanates, and it stops as soon as the user releases the slicing plane. The line plot representing the delta TEC values was paired with a 'triangle wave sound' due to its spatialization properties. The pitch of the sound was mapped to the value of the time-series line plot, changing in proportion to the delta TEC value reported at the timestamp

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Table 1: This table lists the sounds we used in the study, when they were used, and what they indicated to the user.

	Sound Design	Data	Object	Auditory Stimulus	Parameter Mapping
1	Context-based (Task 1)	Storm intensity (Minimum SYM-H value)	Solar activity level plot	Fire crackling sound	Amplitude modulation
2	Action-based (Task 2)	Delta TEC data shown in Figure 1 right-side	Slicing plane	Triangle wave	Pitch modulation
3	Event-based (Task 2)	Sunrise terminator across the delta TEC data	Slicing plane	Rooster sound	NA
4	Event-based (Task 2)	Sunset terminator across the delta TEC data	Slicing plane	Sound of cicadas	NA

as the user sweeps the slicing plane along the time axis. The frequency range for this mapping was chosen between 110 Hz and 440 Hz.

3. **Context-based audio:** The context-based audio represented solar activity intensity via fire-crackling sound. The recorded sound was downloaded from the *freesound.org* website. The sound sample was converted from stereo to mono, the fade-in fade-out effects were applied to the same, and it was normalized to 0 dB using *Audacity*, an open-source digital audio editor. We added another dimension to it by **mapping its amplitude** to the intensity of the storm (the minimum SYM-H value). The signal amplitude of the audio changed from one storm to another based on its minimum SYM-H value. A linear mapping was done between the least minimum SYM-H value and the highest minimum SYM-H value, ranging from -234 to -109, to the signal amplitude values 0.1 to 1.

2.2 Participants

Fifty participants analyzed nine solar storms, completing two tasks:
Task 1: Identifying patterns in solar activity and storm intensity
Task 2: Associating sunrise/sunset with delta TEC values

Performance was assessed via task metrics (e.g., trends identified, response accuracy), NASA Task Load Index, self-reported level of confidence, and System Usability Scale (SUS) for user preferences. The study was approved by the organization's Institutional Review Board (IRB). The design of the prototype immersive analytics application was evaluated separately at a space weather conference with 42 space science experts [3].

3 RESULTS

3.1 Findings based on the sound design

Event-based audio: Event-based sonification used in Task 2 proved to be most helpful in finding patterns and relations in complex datasets. We identified a phenomenon suggesting that the event-based sonification helped the users with temporal orientation in the data space. Whenever audio marks an event or a point in time, users are able to make stronger associations with audio and find correlations. In our study, the audio represented sunrise and sunset points during a storm day, and users were able to find correlations between the delta TEC variable and the sunrise/sunset times. The impact of event-based sonification in an immersive analytics environment is the strongest finding of our study.

Action-based audio: The association between actions and sounds is essential to human perception. The use of audio attached to the slicing plane movement worked favorably, as reported in our previous experiment as well [4, 2]. In our study, users made associations between audio and action intuitively, and therefore, we

recommend to include audio with actions in design for immersive analytics applications.

Context-based audio: The fire crackling sound was mapped spatially to the solar activity plot in Task 1 for context, with the loudness of the audio mapped to the intensity of the storm. Some participants mentioned that it provided an additional dimension and calmness while doing the study. However, the mean values of the NASA TLX results did not show any significant difference from the visual-only scenario. Therefore, adding context-based background audio to the developed immersive scenarios is up to user preferences.

3.2 Quantitative findings based on the number of patterns identified, the level of confidence, user task responses, and the NASA-TLX and SUS questionnaires

1. **Pattern Recognition:** Event-based audio facilitated temporal orientation. Audio-visual participants identified 70% more trends, such as associations between delta TEC extrema and time of day (e.g., "maxima before sunrise, minima before sunset") than the visual-only scenarios.
2. **Confidence Level:** Slightly higher confidence was reported in audio-visual conditions during Task 2, where pitch-based mapping and event-based sonification were used, though it was not statistically significant.
3. **Task Load:** Audio-visual scenarios did not significantly reduce workload compared to visual-only setups.
4. **Usability:** The audio-visual condition scored a SUS of 70.3 (above average), indicating user preference.

4 CONCLUSION

In this work, we evaluated the impact of sonification in an immersive analytics environment by performing a user study with 50 participants. We measured the impact of sonification by comparing audio-visual and visual-only scenarios based on the number of patterns identified by users, the level of confidence in each scenario, user task responses, and NASA-TLX and SUS questionnaires with 50 participants.

This study underscores the potential of sonification in enhancing immersive analytics. While context-based audio added minimal benefits, event-based sonification notably improved temporal pattern recognition. These findings inform future designs for immersive analytics in mixed-reality environments, emphasizing the integration of meaningful audio cues to augment analytical tasks. Further research should examine the effects of spatial versus non-spatial audio and varied spatial configurations.

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